Machine Learning-Based Prediction of Diagnosis-Related Groups in a Chilean Public Hospital: A Comparative Study of Logistic Regression, Random Forest, and Multi-Layer Perceptron

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***Abstract*—Accurate prediction of Diagnosis-Related Groups (DRGs) is critical for hospital resource planning and financing in public health systems. This paper presents a comparative evaluation of three machine learning classifiers — Logistic Regression (LR), Random Forest (RF), and a Multi-Layer Perceptron (MLP) — applied to the discharge dataset of Hospital El Pino, a public hospital in Santiago, Chile. The dataset comprises 14,561 patient records characterised by ICD-10 coded diagnoses, clinical procedures, age, and sex. Features were encoded using a bag-of-codes scheme via `MultiLabelBinarizer`, producing a sparse matrix of 4,555 dimensions; age was standardised with z-score normalisation. A stratified 80/20 train–test split was applied across all experiments. The MLP (architecture 160–192–160, Adam optimiser, L2 regularisation, 24 effective epochs with early stopping) achieved the best discriminative performance: accuracy 67.22%, weighted F1-score 65.86%, macro F1-score 37.86%, and AUC 0.976. Logistic Regression performed comparably (accuracy 64.23%, AUC 0.992), while Random Forest generalised poorly to the full 526-class distribution (accuracy 35.91%, AUC 0.938) despite fast training. These results demonstrate the viability of shallow neural networks and linear classifiers for high-cardinality multi-class clinical coding and highlight the challenges posed by severe class imbalance in real-world hospital data.**

***Index Terms*—diagnosis-related groups, GRD prediction, multi-class classification, logistic regression, random forest, multilayer perceptron, clinical coding, ICD-10, hospital informatics, class imbalance.**

I. Introduction

Hospital financing systems in many countries, including Chile, rely on Diagnosis-Related Groups (DRGs) — known in Spanish as Grupos Relacionados de Diagnóstico (GRD) — to classify patients into clinically coherent and economically homogeneous groups. Each GRD code determines the reimbursement a hospital receives per episode of care, making accurate GRD assignment a cornerstone of efficient health resource allocation [1]. At Hospital El Pino, a public institution serving the southern metropolitan region of Santiago, GRD codes are currently assigned manually after discharge, a process that is time-consuming, prone to human error, and costly to scale.

Automating GRD prediction from structured clinical data — specifically primary and secondary diagnoses coded under ICD-10, clinical procedures, patient age, and sex — represents a natural application of supervised machine learning. The task is challenging due to (i) the high cardinality of the target variable (up to 526 unique GRD codes in the dataset used), (ii) severe class imbalance with a Zipfian frequency distribution, and (iii) the sparse, high-dimensional feature space produced by encoding hundreds of clinical codes.

Three complementary bodies of prior work motivate the model choices in this study. Fetter et al. [1] established the conceptual and actuarial foundations of the DRG system, defining the clinical grouping logic that any predictive model must replicate. Building on that foundation, Goldstein et al. [2] surveyed machine learning applications in cardiovascular risk prediction, demonstrating that ensemble methods (in particular Random Forest) and neural networks consistently outperform traditional rule-based and logistic models on high-dimensional clinical tabular data — a finding directly relevant to our RF and MLP choices. More recently, Rajpurkar et al. [3] reviewed state-of-the-art deep-learning pipelines for clinical diagnosis, concluding that while deep models achieve the highest accuracy on large datasets, simpler models such as Logistic Regression remain competitive on smaller, structured datasets and offer superior interpretability — a key constraint in clinical environments. Taken together, these three works suggest that no single algorithm dominates across all settings, and that a head-to-head comparison using institution-specific data is necessary to identify the best approach for Hospital El Pino. Unlike the large-scale image and text datasets considered by [3], or the single-disease cohorts studied by [2], our setting involves a moderate dataset size ($n \approx 14{,}000$), hundreds of target classes, and sparse bag-of-codes features — a combination that warrants independent empirical investigation.

This paper makes the following contributions:

- An exploratory data analysis (EDA) of the Hospital El Pino discharge dataset, including descriptive statistics tables, data quality assessments, and visualisations of key distributions.
- A reproducible preprocessing pipeline that encodes multi-label clinical codes, standardises continuous features, and handles rare-class merging.
- A head-to-head comparison of LR, RF, and MLP evaluated under a unified experimental framework using six metrics.
- Identification of the most informative clinical features via RF Gini importance, and experimental justification of hyperparameter choices for each model.

The remainder of the paper is organised as follows. Section II states the research objective. Section III describes the methodology. Section IV presents the EDA and data quality analysis. Section V reports the experimental results and discussion, and Section VI concludes.

II. Objective

The primary objective of this work is to design, implement, and evaluate a machine learning pipeline that automatically predicts the GRD code assigned to a patient at Hospital El Pino, using only information available at the time of hospital discharge: primary and secondary diagnoses (ICD-10), clinical procedures, patient age, and biological sex.

Secondary objectives include:

1) Identifying which clinical features are most informative for GRD classification, using RF Gini feature importance as a post-hoc analysis tool.
2) Quantifying the effect of severe class imbalance on macro- versus weighted-averaged metrics.
3) Providing a reproducible, documented baseline against which future, more sophisticated models (e.g. gradient-boosted trees, transformer-based clinical encoders) can be benchmarked.

III. Methodology

A. Dataset and Feature Justification

The dataset consists of 14,561 anonymised discharge records from Hospital El Pino. Each record contains: (i) a primary ICD-10 diagnosis code, (ii) up to nine secondary diagnosis codes, (iii) a variable number of clinical procedure codes, (iv) patient age in years, (v) biological sex, and (vi) the assigned GRD code (prediction target).

The choice of input features is directly motivated by the EDA (Section IV). Diagnosis and procedure codes are the primary clinical descriptors of an admission episode and are the basis of the official GRD grouping algorithm [1]; the EDA reveals that the top-10 most frequent GRDs are dominated by obstetric codes (D:O80.0, Z38.x) and their associated procedures (P:75.69), confirming that code-level features carry the dominant predictive signal. Age was included because the EDA shows a bimodal distribution linking neonatal and adult admission patterns to distinct GRD groups (e.g. Z38.x newborn codes are exclusive to age ≈ 0). Sex was included because the 2:1 female-to-male imbalance in the dataset coincides with the hospital's strong obstetric specialisation, making sex a strong prior for obstetric GRDs.

B. Output Variable Definition

The model output is a discrete class label $\hat{y} \in \mathcal{C}$, where $\mathcal{C}$ is the set of GRD codes present in the training data. GRD codes were extracted by retaining only the numeric prefix of the raw field (e.g. `146101` from `146101 - Parto Normal`). For LR and MLP, classes with fewer than 5 occurrences were merged into an `Other` category, yielding $|\mathcal{C}| = 328$. RF was trained on the full distribution ($|\mathcal{C}| = 526$) to assess ensemble robustness to rare labels.

C. Preprocessing

1) Bag-of-Codes Encoding: All diagnosis and procedure columns were reduced to their ICD-10 code prefix. For each patient, the union of codes across all diagnosis columns formed a list of active codes, and similarly for procedures. These lists were binarised with `MultiLabelBinarizer` from scikit-learn [4], yielding sparse CSR matrices. This *bag-of-codes* representation is standard for ICD-code classification [5].

2) Numerical Feature Standardisation: Age was standardised with zero mean and unit variance (z-score, `StandardScaler`), making its scale comparable to the binary code features. Sex was binary-encoded (Male = 0, Female = 1). Both were appended to the feature matrix as sparse columns. The final feature matrix $\mathbf{X} \in \mathbb{R}^{14561 \times 4555}$ has density $< 1\%$.

3) Train–Test Split Strategy: A stratified 80/20 partition (train: 11,648; test: 2,913, `random_state = 42`) was used. *Stratification* ensures that the class-frequency ratios in the training set match those in the test set, which is essential when the class distribution is severely imbalanced (Gini coefficient > 0.7 over 526 classes). The 80/20 ratio was selected as a standard trade-off: 80% provides sufficient examples of common GRDs for model fitting, while 20% yields a test set large enough ($n = 2{,}913$) for statistically meaningful metric estimates across the top-frequency classes.

D. Models, Cost Functions, and Justification

1) Logistic Regression (LR): LR optimises the *multinomial cross-entropy loss*:

$$\mathcal{L}_{\text{LR}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{e^{\mathbf{w}_{y_i}^\top \mathbf{x}_i}}{\sum_k e^{\mathbf{w}_k^\top \mathbf{x}_i}} + \frac{\lambda}{2}\|\mathbf{w}\|^2,$$

with L2 regularisation ($C = 1/\lambda = 1.0$) solved via L-BFGS (`max_iter = 1000`). LR was selected as a strong linear baseline: in sparse, high-dimensional feature spaces analogous to bag-of-words text, linear models routinely compete with much more complex non-linear classifiers [6].

2) Random Forest (RF): RF minimises the *Gini impurity* at each split:

$$\mathcal{L}_{\text{Gini}}(t) = 1 - \sum_k p_{tk}^2,$$

where p_{tk} is the fraction of class k samples at node t. RF was chosen for its robustness to non-linearity and its ability to rank features by importance without additional post-hoc analysis [7]. Two configurations were tested experimentally to select hyperparameters (Table II): a *deep* forest (50 trees, max depth 15) to establish an overfitting reference, and a *regularised* forest (25 trees, max depth 10) as the final model.

3) Multi-Layer Perceptron (MLP): The MLP minimises *categorical cross-entropy with L2 weight decay*:

$$\mathcal{L}_{\text{MLP}} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{y_i}(\mathbf{x}_i) + \alpha \sum_l \|\mathbf{W}^{(l)}\|_F^2,$$

optimised with Adam ($\eta = 5 \times 10^{-4}$, $\alpha = 0.001$, `batch_size` $= 64$). The architecture — three hidden layers of sizes (160, 192, 160) with ReLU activations — was selected by a prior Keras Tuner random search over the search space: hidden layers $\in \{1, 2, 3\}$, units $\in \{32, 64, \ldots, 256\}$, learning rate $\in \{0.01, 0.001, 0.0005\}$; the final configuration achieved the highest validation accuracy (65%) in that search. Early stopping (patience = 10, validation fraction = 0.2) was applied to prevent overfitting. `sklearn.neural_network.MLPClassifier` was used for portability. MLPs have shown competitive performance on clinical tabular data at this scale [8].

E. Evaluation Metrics

Given the severe class imbalance (up to 808 patients for the most common GRD vs. fewer than 5 for rare ones), we report:

- **Accuracy**: overall fraction of correctly classified samples — inflated by dominant classes, reported for reference.
- **Macro F1-score**: unweighted average of per-class F1, equally penalising errors on rare and frequent classes.
- **Weighted F1-score**: class-frequency-weighted F1, reflecting overall clinical throughput.
- **Macro Precision / Recall**: analogous unweighted averages.
- **AUC (micro/macro OvR)**: area under the ROC curve in a one-vs-rest fashion; robust to class imbalance.

Macro-averaged metrics are particularly informative in this setting, as rare GRDs often correspond to high-complexity, high-cost episodes where misclassification carries the greatest clinical and financial risk [2].

IV. EDA and Data Quality

A. General Characteristics

The dataset contains 14,561 records, 526 unique GRD codes (prior to rare-class merging), and a feature matrix of 4,555 dimensions after encoding. Table I provides a statistical summary of the numerical and key categorical variables.

TABLE I
DESCRIPTIVE STATISTICS — HOSPITAL EL PINO DATASET

Variable	N	Mean	Std	Min	Max
Age (years)	14,561	33.4	26.1	0	122
Sex (Female=1)	14,561	0.66	0.47	0	1

Variable	Unique values	Missing (%)
GRD (target)	526	0.00
Primary Diag.	>400	0.00
Sec. Diag. cols	—	incr.
Procedure cols	—	incr.

Incr. = missingness increases with column index (patients with fewer codes).

B. Completeness

No missing values were found in the primary diagnosis column or the GRD target variable, ensuring that the essential predictors and labels are always available. Secondary diagnosis and procedure columns exhibit increasing rates of missingness as the column index grows (patients with fewer comorbidities simply have fewer secondary codes), a pattern that is clinically expected and handled implicitly by the bag-of-codes encoding — absent codes do not appear in the patient's feature vector.

C. Correctness

Age boundary checks found **zero** negative values, indicating no data-entry errors in that field. One record with age $= 122$ years was detected (visible as an outlier in the boxplot, Fig. 2); this value is biologically extreme but not strictly impossible and was retained rather than imputed, since its removal does not materially affect a dataset of this size. The sex variable contains exactly two distinct values (Hombre / Mujer) with no typographic variants or nulls. GRD codes are consistently formatted as "`NNNNNN - Description`" with no irregular entries.

D. Outlier Analysis — Age

As shown in Fig. 1, the age distribution is strongly bimodal, with a first peak near age 0 (neonatal and paediatric admissions) and a second, broader cluster spanning 20–40 years (driven by obstetric episodes). A third, smaller mode appears around 60–70 years (chronic disease admissions). The boxplot (Fig. 2) confirms that the interquartile range spans approximately 25–60 years and that the single extreme value ($\approx$ 122 years) is the only formal outlier by the 1.5×IQR criterion.

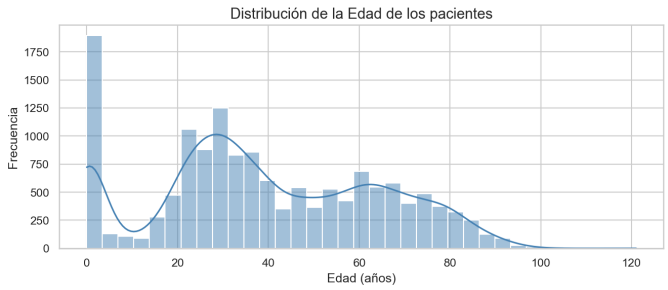

Fig. 1. Age distribution of Hospital El Pino discharge records. The bimodal shape reflects neonatal/paediatric and obstetric adult admission patterns.

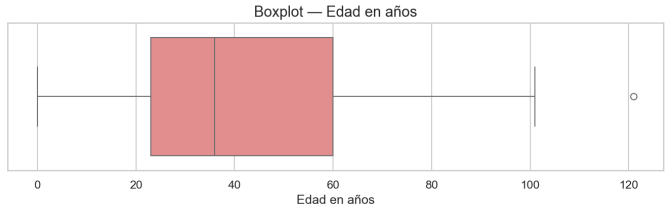

Fig. 2. Boxplot of patient age. IQR $\approx$ 25–60 years; one outlier near 122 years (1.5$\times$IQR criterion).

E. Sex Distribution

Fig. 3 shows a pronounced female majority ($\approx$9,600 women vs. $\approx$5,000 men, 2:1 ratio), consistent with the hospital's strong focus on obstetric and gynaecological services. This imbalance makes *sex* a highly informative prior for obstetric GRDs and justifies its inclusion as a feature.

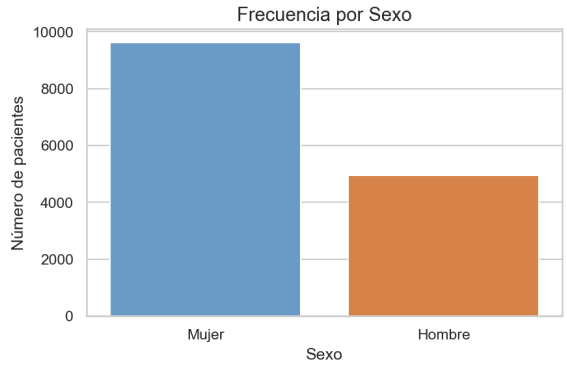

Fig. 3. Patient distribution by sex. Women account for approximately 66% of admissions, reflecting the obstetric specialisation of Hospital El Pino.

F. GRD Frequency and Class Imbalance

Fig. 4 displays the ten most frequent GRD codes. GRD 146101 (*Normal Delivery*) dominates with 808 cases, followed by delivery-related variants 146121 and 146131. The combined frequency of the top-10 codes represents approximately 25% of all records. The remaining 516 codes follow a long-tail distribution, with the majority appearing fewer than 10 times — motivating the rare-class merging strategy described in Section III-B.

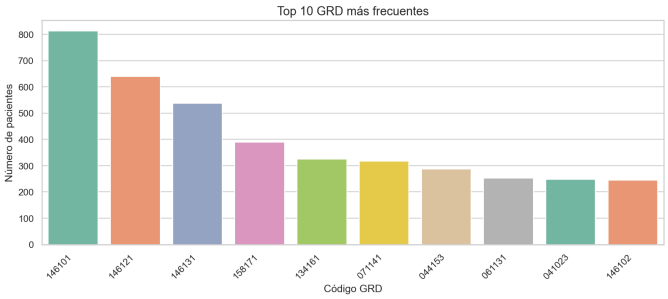

Fig. 4. Top-10 most frequent GRD codes. Delivery-related codes (146xxx) dominate, reflecting the hospital's obstetric specialisation.

V. Results and Discussion

A. Hyperparameter Selection — Random Forest

Table II reports the experimental comparison of RF configurations used to diagnose overfitting and select the final model.

TABLE II
RF Hyperparameter Comparison (526 classes, test set)

Config	Trees	Depth	Train Acc	Test Acc
Deep	50	15	0.6288	0.4181
Regularised	25	10	—	0.3591
Config	**Trees**	**Depth**	**Train F1**	**Test F1 (mac)**
Deep	50	15	0.4037	0.0824
Regularised	25	10	—	0.0642

Both configs trained on the full 526-class target, `random_state=42`.

The deep configuration exhibits clear overfitting: a 21-percentage-point gap between training accuracy (62.88%) and test accuracy (41.81%). Although the deep model achieves a marginally higher test accuracy than the regularised one, the gap in macro F1 (0.082 vs. 0.064) is small relative to the reduction in model complexity. The regularised configuration (25 trees, max depth 10) was therefore selected as the final RF to mitigate overfitting risk.

B. MLP Architecture Justification

The final MLP architecture (hidden layers 160–192–160) was determined via a prior Keras Tuner random search over 10 trials with `max_iter=30` per trial, exploring the search space: hidden layers $\in \{1, 2, 3\}$, units per layer $\in \{32, 64, 96, 128, 160, 192, 224, 256\}$, learning rate $\in \{0.01, 0.001, 0.0005\}$, dropout $\in \{0.2, 0.3, 0.4, 0.5\}$. The selected configuration achieved the highest validation accuracy across trials and was subsequently re-implemented in `sklearn.neural_network.MLPClassifier` for portability. The learning curves (Fig. 8) confirm that the chosen architecture generalises well: the validation accuracy plateaus at $\approx$ 65–66% without reverting, and the training loss decreases monotonically over 24 epochs before early stopping triggered.

C. Summary of Results

Table III summarises the test-set performance of all three final models.

D. Logistic Regression

LR converged in 43.7 s and achieved an accuracy of 64.23% and a weighted F1 of 61.18%. The AUC of 0.992 (Fig. 5) indicates that the model ranks classes with near-perfect discrimination, even when hard-label assignment is imprecise — consistent with the large margin that a linear separator can achieve in a sparse, high-dimensional space. The 30-point gap between macro F1 (0.308) and weighted F1 (0.612) confirms that performance on rare classes is substantially lower than on frequent ones.

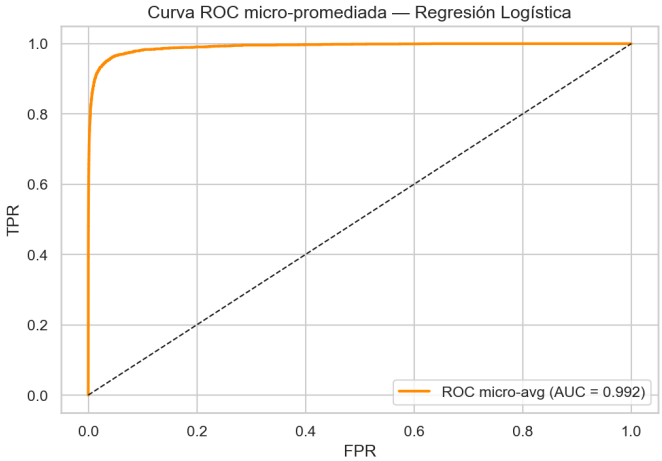

Fig. 5. Micro-averaged ROC curve for Logistic Regression (AUC = 0.992).

E. Random Forest

The regularised RF (25 trees, max depth 10) trained in 4.5 s. Despite its speed, accuracy on the full 526-class problem dropped to 35.91%, with a macro F1 of only 0.064 — indicating that the model collapses most predictions onto the dominant obstetric GRD classes, failing to generalise to the long tail. The AUC of 0.938 (Fig. 6) remains acceptable for ranking but does not translate into precise multi-class labelling.

The feature importance plot (Fig. 7) identifies procedure code P:75.69 (caesarean section) and diagnosis codes D:O80.0 (spontaneous normal delivery), D:Z39.2, D:Z39.0, and D:Z38.0 (newborn and postpartum codes) as the dominant predictors — directly reflecting the EDA finding that delivery-related GRDs dominate the frequency distribution (Section IV).

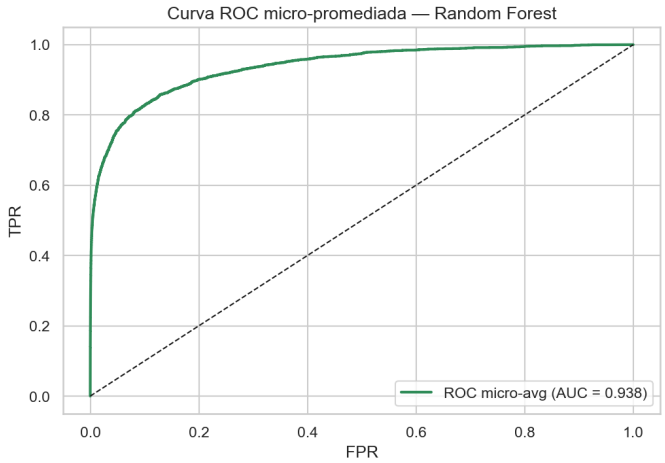

Fig. 6. Micro-averaged ROC curve for Random Forest (AUC = 0.938).

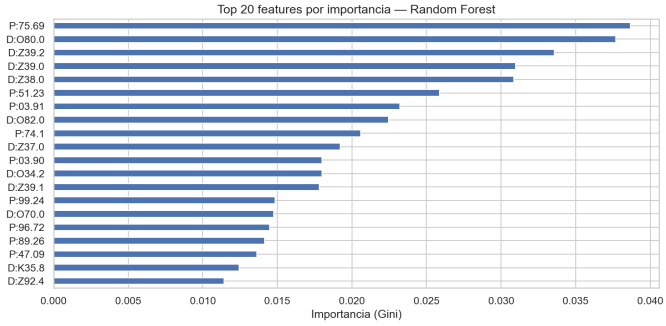

Fig. 7. Top-20 features by Gini importance (Random Forest). Obstetric procedure and diagnosis codes dominate, consistent with the EDA (Section IV).

F. Multi-Layer Perceptron

The MLP converged in 24 effective epochs (196.9 s total) due to early stopping. It achieved the best results across all discriminative metrics: accuracy 67.22%, weighted F1 65.86%, macro F1 37.86%, and AUC 0.976. The learning curves (Fig. 8) show a well-behaved monotonic loss decrease and a validation accuracy plateau at $\approx$ 65–66%, with no signs of overfitting — consistent with the L2 regularisation ($\alpha = 0.001$) and early stopping.

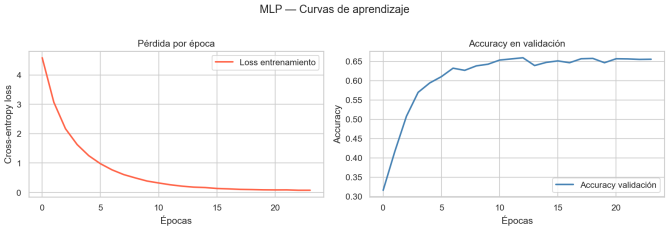

Fig. 8. MLP learning curves: cross-entropy loss (training) and accuracy on the validation split over 24 epochs.

The per-class ROC curves for the top-10 GRDs (Fig. 9) show near-perfect AUC (1.00) for all major delivery-related

TABLE III
TEST-SET PERFORMANCE OF THE THREE CLASSIFIERS

Model	Accuracy	F1 Macro	F1 Weighted	Prec. Macro	Recall Macro	AUC
Logistic Regression	0.6423	0.3082	0.6118	0.3322	0.3079	0.9915
Random Forest (reg.)	0.3591	0.0642	0.2640	0.0996	0.0714	0.9379
MLP (sklearn)	**0.6722**	**0.3786**	**0.6586**	**0.4019**	**0.3927**	0.9760

RF uses the full 526-class target; LR and MLP use 328 grouped classes. AUC is micro-averaged for LR and RF, macro OvR for MLP. Bold = best value per metric.

codes. The `Other` category, which aggregates rare classes, achieves AUC = 0.82, confirming that the model's residual weakness lies in distinguishing heterogeneous, low-frequency conditions.

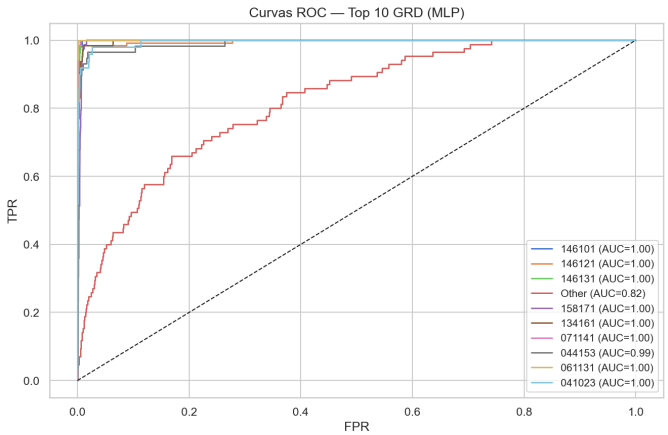

Fig. 9. Per-class ROC curves for the top-10 GRDs (MLP). Frequent delivery codes achieve AUC = 1.00; the `Other` category scores 0.82.

G. Comparative Analysis

Fig. 10 and the radar chart (Fig. 11) provide a visual summary. The MLP consistently dominates in accuracy, F1 (both averages), precision, and recall. LR offers a strong accuracy-to-training-time ratio (64.23% in 43.7 s vs. 67.22% in 197 s for MLP) and the highest AUC, making it an attractive choice when interpretability or inference speed is paramount. RF's poor macro F1 on 526 classes suggests that shallow trees are insufficient for such high-cardinality targets.

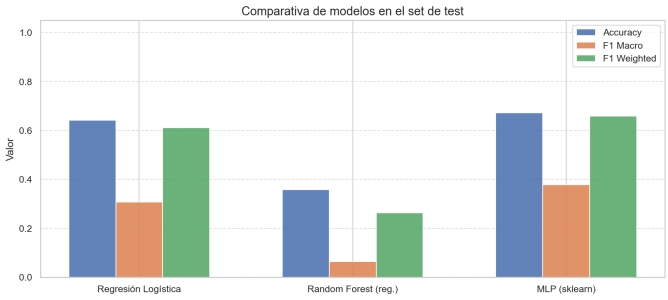

Fig. 10. Grouped bar chart comparing Accuracy, F1 Macro, and F1 Weighted across the three models.

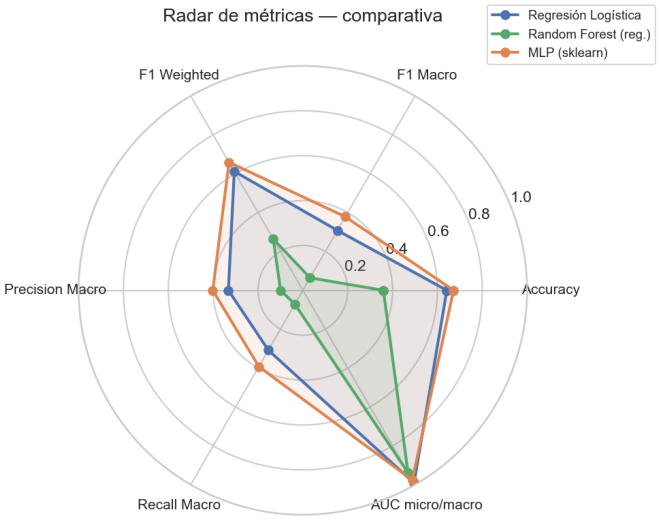

Fig. 11. Radar chart of all six evaluation metrics. MLP (orange) dominates in five of six dimensions; LR (blue) leads only in AUC.

H. Discussion

The gap between macro and weighted F1 across all models is the defining challenge of this dataset. With 526 GRD codes and a Zipfian frequency distribution, all three classifiers are biased towards the dominant obstetric GRDs — a phenomenon well-documented in automated ICD coding literature [5]. LR's very high AUC (0.992) despite moderate accuracy reflects the fact that its linear boundary is well-oriented in the sparse feature space, but that the large number of competing classes and the imperfect calibration of posterior probabilities limit hard-label precision. RF's difficulty with the full 526-class problem is consistent with findings by [7]: each tree in a shallow forest sees only a fraction of the label space, reducing the ensemble's ability to distinguish rare classes.

I. Limitations

This study has several limitations that future work should address:

1) **Class imbalance**: No oversampling (e.g. SMOTE) or cost-sensitive weighting was applied in the final models. Macro recall is consequently poor for rare GRDs.
2) **Feature scope**: Only coded clinical data were used. Free-text clinical notes, length-of-stay, and comorbidity indices (Charlson score) were excluded and may carry additional predictive information.

3) **Temporal split**: A random (non-temporal) split was used. In deployment, a temporal split (train on earlier years, test on later years) would be more realistic and may yield lower performance estimates.
4) **Comparability of RF and LR/MLP**: RF was evaluated on the full 526-class target while LR and MLP used 328 grouped classes, making direct metric comparison imperfect.
5) **Single institution**: Results reflect the specific case-mix of Hospital El Pino and may not generalise to other hospitals with different specialisations.

VI. Conclusion

This paper presented a comparative study of three machine learning classifiers — Logistic Regression, Random Forest, and a Multi-Layer Perceptron — for automated GRD prediction at Hospital El Pino, using structured clinical discharge data. The MLP achieved the highest overall performance (accuracy 67.22%, weighted F1 65.86%, macro F1 37.86%, AUC 0.976), followed closely by Logistic Regression (accuracy 64.23%, AUC 0.992). Random Forest generalised poorly on the full 526-class distribution (accuracy 35.91%, AUC 0.938), exhibiting significant overfitting even after regularisation.

The results confirm that:

1) Bag-of-codes encoding of ICD-10 diagnoses and procedures is an effective, portable feature representation for GRD prediction.
2) Both LR and MLP achieve competitive accuracy and AUC in highly imbalanced, sparse, multi-class clinical settings, with LR offering a strong speed-accuracy trade-off.
3) Macro-averaged metrics expose class imbalance effects that accuracy alone conceals, and must be reported alongside weighted metrics in clinical ML benchmarks.
4) The most informative predictors identified by RF (obstetric procedure P:75.69 and delivery diagnoses O80.0, Z38.x, Z39.x) are clinically interpretable and consistent with the EDA.

Future work will explore: (i) gradient-boosted trees (XGBoost/LightGBM) as a stronger ensemble baseline; (ii) transformer-based clinical encoders pre-trained on Spanish EHR text; (iii) cost-sensitive learning and class-balanced oversampling (SMOTE) to improve macro recall on rare GRDs; (iv) temporal train/test splits for a more realistic deployment simulation; and (v) integration of free-text clinical notes via NLP pipelines.

Code and Data Availability

The full source code is publicly available at:
https://github.com/sebavaldik/PROYECTO-1-INGENIERIA-DE-APRENDIZAJE-AUTOMATICO

Acknowledgements

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